

SERVICE MANUAL

MRI Devices Corporation
3545 SW 47th Avenue
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Document TR0255S

Revision C

GE Signa Excite 3.0T High Resolution Brain Array

GE Part Number: M1385AS

MRIDC Part Number: 800255



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Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

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- Call 1-800-548-3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

Language Policy For Service Documentation (Dir. 2128126)

W A R N I N G

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
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- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

This is a service manual for the GE SIGNA EXCITE 3.0T High Resolution Brain Array.



SHIPPING LIST – TABLE 1-1

Description	GE Part #	MRIDC #	Qty
3.0T 8 Ch HiRes Brain Array / Baseplate Assm.	2380637-2	102859	1
Phantom Positioner	2317112-3	102558	1
Operator Manual	2380637-3	500111	1
Service Manual	2380637-4	TR0255S	1
Thick Head Pad	2317112-9	102985	1
Thin Head Pad	2317112-10	102984	1
Spacer Pad	2317112-7	101622	1
Wedge Pad	2317115-8	101407	2

1-2 Compatibility

This coil is compatible with the following hardware configurations

- GE SIGNA EXCITE 3.0T (8-channel)

1-3 Related Documentation

- 3.0T EXCITE SERVICE METHODS CD-ROM Part #2389160-1
- 3.0T 8 Channel High Resolution Brain Array Operator's Manual, 2380637-3

1-4 Environmental Requirements

Storage Requirements

Coil should be stored flat in the scanner room.

Dimensions

Coil Dimension:	417.6 mm x 393.4 mm x 406.4 mm	(16.44 in x 15.49 in x 16.00 in)
Phantom Positioner:	225.6 mm x 228.6 mm x 23.9 mm	(8.88 in x 9.00 in x 0.94 in)

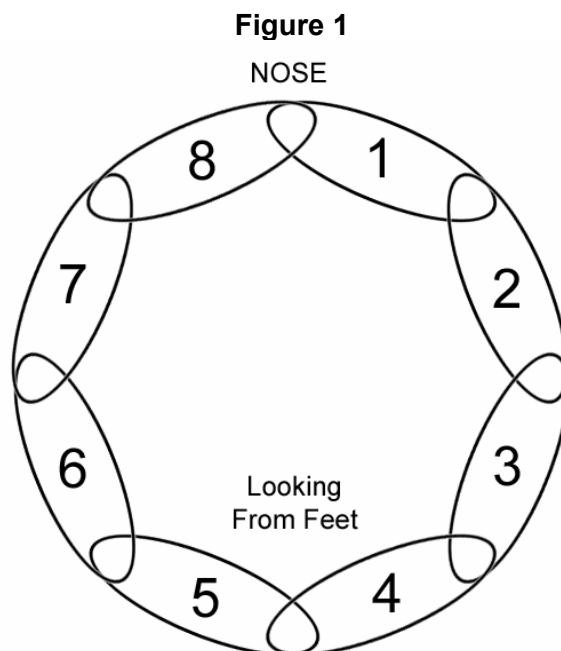
Weight

Coil Weight:	7.20 Kg	(16.0 lb)
Phantom Positioner:	0.45 Kg	(1.0 lb)

1-5 Theory of Operation

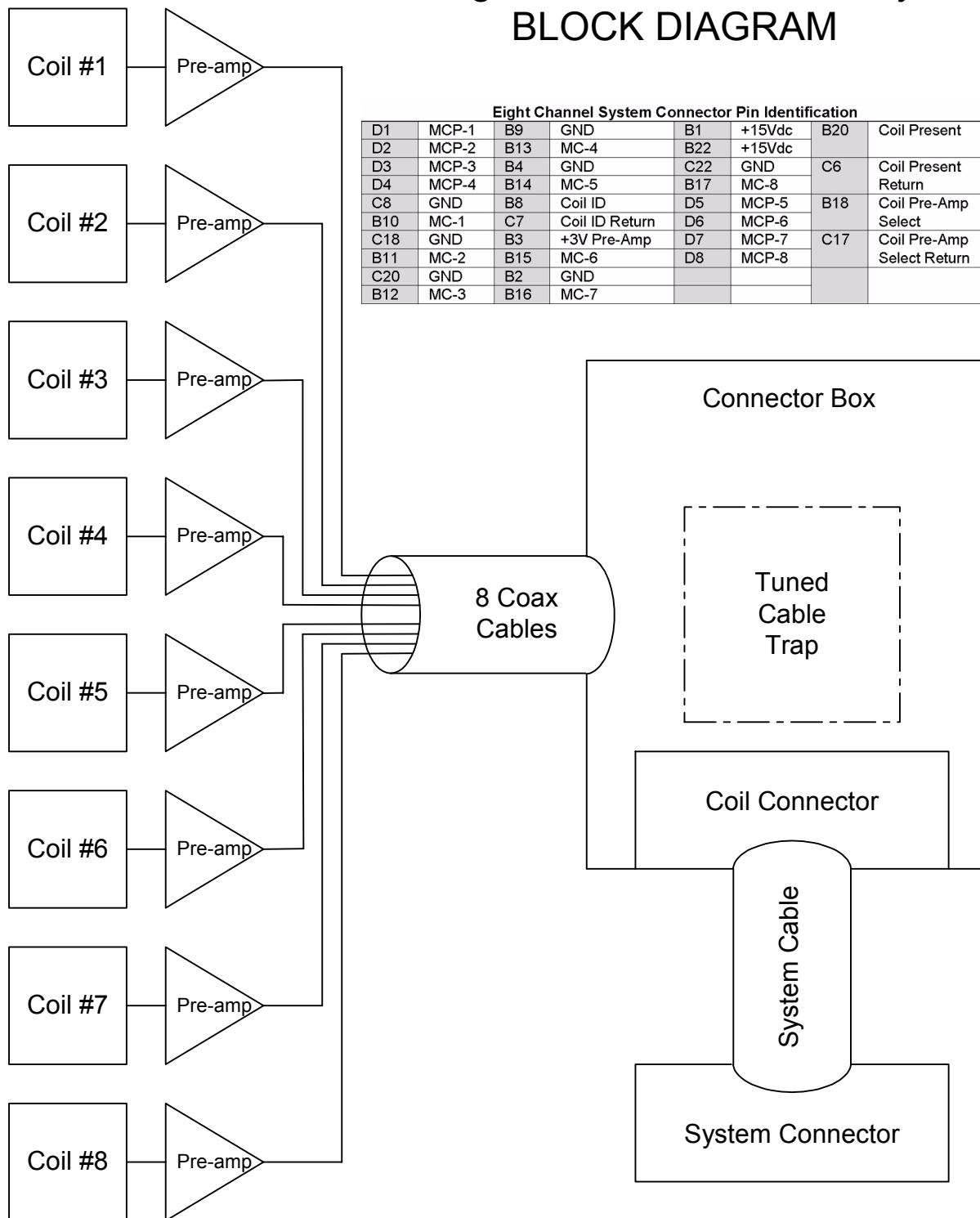
Refer to Figure 1 below and the Block Diagram on the following page.

The 3.0T 8 Channel High Resolution Brain Array consists of 8 elements. Each element is comprised of a loop arranged azimuthally around the head. Each loop is approximately 45° in azimuthal extent. The center of Loop #1 is located 22.5° rotated in the clockwise direction (as viewed from the feet) with respect to the nose. Each of the next seven loops is rotated clockwise by 45° successively. Loops #1 and #8 are located anteriorly while loops #4 and #5 are located posteriorly. Loops #2 and #3 are located on the patient Left side while Loops #6 and #7 are located on the patient Right side. During transmit, each loop is decoupled with both passive and active decoupling, with an additional active preamp protection network. During receive, signals from 1 – 8 are directed to separate ultra-low-input impedance preamp modules placed directly at the output of the loop, providing additional element decoupling.



Loop Element Diagram

GE SIGNA EXCITE 3.0T High Resolution Brain Array BLOCK DIAGRAM



SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None

2-1-2 Installing the Coil

The name for this coil is: **8HRBRAIN**

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3-2, Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2, Coil Imaging Performance Verification, and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	MRIDC #	Qty
18cm Diameter Spherical TLT Phantom	46-265826G6	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2-2 Data Collection

{See external addendum for QA procedure}

3-2-3 Data Analysis

[See external addendum for QA procedure]

3-2-4 Test Explanation and Results Interpretation

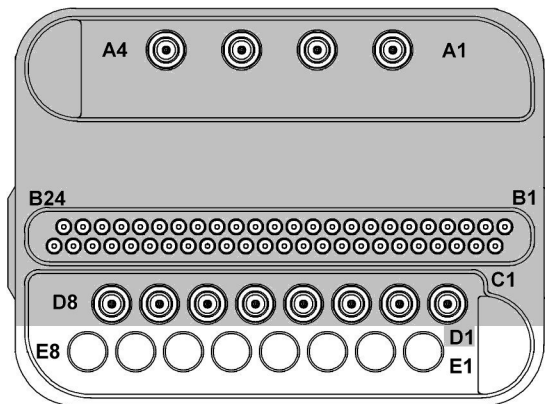
[See external addendum for QA procedure]

3-3 External Cable Check

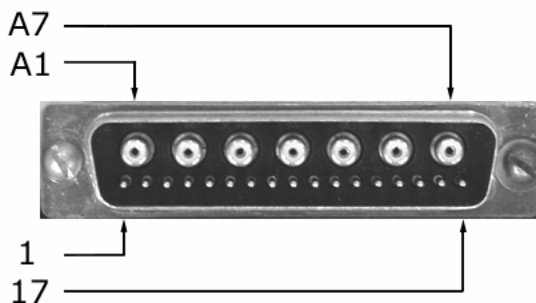
Each MCP signal connection has a DC blocking capacitor in the cable connector. Therefore, continuity of the RF signal conductors cannot be determined in the usual way, by measuring from pin to pin using a multimeter. The external cable check must be performed in two parts, section 3-3-1 for the RF signal conductors and 3-3-2 for the PIN diode control conductors.

Use Figure 3-3 for pin identification while performing the procedures in sections 3-3 and 3-4.

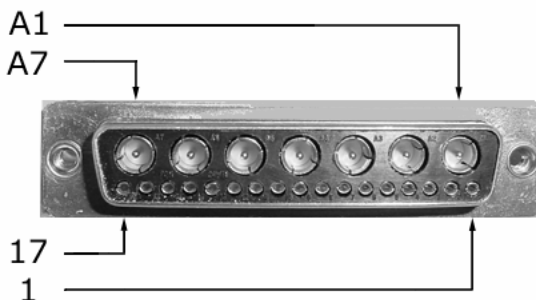
System and Coil Connectors Pin Identification Figure 3-3
Hypertronics System Connector



SubD Coil Cable Connector



SubD Coil Connector



3-3-1 RF Signal Channels

The following procedure will rule out a short to GND for each of the 16 signal pins detailed in Table 3-3-1. This check does not, however, determine continuity for each of the MCP signal connections.

Check the coil cable using the following procedure. Referring to Section 5-2, External Cable Replacement, disconnect the SubD coil cable connector. Using a digital multimeter, measure each of the 16 signal pins designated in Table 3-3-1 with respect to GND. Verify that the signal pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-1

Signal Channel	SubD Coil Cable Connector	GND
MCP1	A7	1,3 and shield
MCP2	A6	
MCP3	A5	
MCP4	A4	
MCP5	A3	
MCP6	A2	
MCP7	2	
MCP8	A1	
Signal Channel	Hypertronics System Connector	GND
MCP1	D1	B2, B4, B9, C8, C18, C20, C22
MCP2	D2	
MCP3	D3	
MCP4	D4	
MCP5	D5	
MCP6	D6	
MCP7	D7	
MCP8	D8	

If one or more of the pins are shorted to GND, replace the coil cable.

3-3-2 PIN Diode Control Channels

With the coil cable disconnected, use a multimeter to determine continuity between the designated pins for each control channel in Table 3-3-2. Also verify that the control pins are not shorted to GND.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3-2

Control Channel	SubD Coil Cable Connector	Hypertronics System Connector
MC 1	17	B10
MC 2	16	B11
MC 3	15	B12
MC 4	14	B13
MC 5	4	B14
MC 6	5	B15
MC 7	6	B16
MC 8	7	B17
GND	1,3 and shield	B2, B4, B9, C8, C18, C20, C22

If one or more of the readings are > 3 ohms, replace the coil cable. If all readings are < 3 ohms, proceed to the **Diode Check**.

3-4 PIN Diodes Check

With the coil cable connected to the coil, use a multimeter to measure the diode drop across the coil connector pins detailed in Table 3-4. Use Figure 3-3 for pin identification while performing this procedure.

PIN DIODE EXPECTED READINGS – TABLE 3-4

Hypertronics System Connector		Reading
Positive Lead	Negative Lead	
B10	GND [Shield]	2.25 ± 0.2
GND [Shield]	B10	Open
B11	GND [Shield]	2.25 ± 0.2
GND [Shield]	B11	Open
B12	GND [Shield]	2.25 ± 0.2
GND [Shield]	B12	Open
B13	GND [Shield]	1.72 ± 0.2
GND [Shield]	B13	Open
B14	GND [Shield]	1.72 ± 0.2
GND [Shield]	B14	Open
B15	GND [Shield]	2.25 ± 0.2
GND [Shield]	B15	Open
B16	GND [Shield]	2.25 ± 0.2
GND [Shield]	B16	Open
B17	GND [Shield]	2.25 ± 0.2
GND [Shield]	B17	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo 3-2 *Coil Imaging Performance Verification*. If the problem still exist, replace the 3.0T 8 Channel High Resolution Brain Array.

3-5 Mechanical Hardware Check

Not Applicable

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.



Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution.

The GE SIGNA EXCITE 3.0T High Resolution Brain Array, Baseplate and Head Pads may be cleaned by wiping with a cloth dampened with a solution of 30% isopropyl alcohol and 70% tap water or a 10% Bleach solution.

4-2 Special Care Requirements

None

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

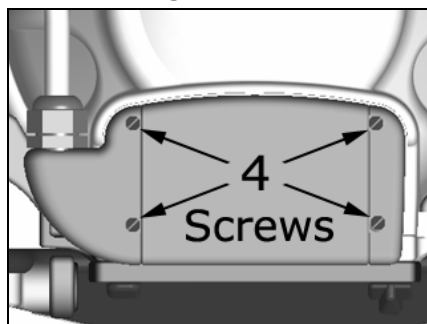
5-1 Disassembly of Coil

Not Applicable

5-2 External Cable Replacement

The 4 coil connector mounting screws are located at the superior end of the coil. Remove the 4 coil connector mounting screws as detailed in Figure 5-2-1. Remove the coil connector by pulling away from the coil in the superior direction.

Figure 5-2-1



Remove 4 Screws

5-3 Mechanical Hardware Replacement

Not Applicable

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1

Description	GE Part #	MRIDC #
3.0T 8 Ch HiRes Brain Array / Baseplate Assm.	2380637-2	102859
Cable	2380637-5	101858
Phantom Positioner	2317112-3	102558

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Description	GE Part #	MRIDC #
Thick Head Pad	2317112-9	102985
Thin Head Pad	2317112-10	102984
Spacer Pad	2317112-7	101622
Wedge Pad	2317115-8	101407

SECTION 7 – APPENDIX

7-1 Schematic

Refer to Block Diagram under Section 1-5 Theory of Operation.

7-2 Coil Configuration

Check the system coil configuration file for the 8HRBRAIN coil configuration. If 8HRBRAIN has not been installed, select 8HRBRAIN from the list of available coils, and install.

Revision History

Rev	Date	Author	Primary Reason for Change
A	07-14-03	L. Hyler	Initial Release
B	09-10-03	L. Hyler	Add Correction from GE E-mail [JK 08-25-03] Updated "Damage in Transportation" Page 2
C	09-26-03	L. Hyler	Point to Appendix from Sections 3-2-2 through 3-2-4